

Arun Durvasula, PhD

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Education

PhD Human Genetics, University of California, Los Angeles	2016-2021
BS Biotechnology, University of California, Davis	2011-2015

Experience

Postdoctoral Research Fellow. Department of Genetics, Harvard Medical School. Mentors: Dr. David Reich & Dr. Alkes Price	2021-Present
Graduate Research Fellow. Department of Human Genetics, UCLA. Mentors: Dr. Kirk Lohmueller & Dr. Sriram Sankararaman	2016-2021
Post-Bac Researcher. Max Planck Institute for Plant Breeding Research. Mentor: Dr. Angela Hancock	2015-2016
Undergraduate Researcher. University of California, Davis. Mentor: Dr. Jeffrey Ross-Ibarra	2013-2015

Publications & Preprints

(* = co-first author, # = co-senior author)

15. Zhang X*, Kim BY*, Singh A, Sankararaman S, **Durvasula A**#, Lohmueller KE#. MaLAdapt reveals novel targets of adaptive introgression from Neanderthals and Denisovans in worldwide human populations. Preprint@bioRxiv (in revision, *Molecular Biology and Evolution*)
14. Huang B, **Durvasula A**, Mousavi N, Ziaei-Jam H, Maksimov M, Lohmueller KE, Gymrek M. Genome-wide selection inference at short tandem repeats. Preprint@bioRxiv (in revision, *American Journal of Human Genetics*)
13. Wei X*, Robles C*, Pazokitoroudi A, Ganna A, Gusev A, **Durvasula A**, Gazal S, Loh PR, Reich D, Sankararaman S. The lingering effects of Neanderthal introgression on human complex traits. Preprint@bioRxiv (in revision, *eLife*)
12. Molloy EK, **Durvasula A**, Sankararaman S. Advancing admixture graph estimation via maximum likelihood network orientation (2021). *Bioinformatics*. (Conference proceeding of ISMB).
11. **Durvasula A**, Lohmueller K. Negative selection on complex traits limits phenotype prediction accuracy between populations (2021). *American Journal of Human Genetics*.
 - F1000 recommended
10. Boocock J, Sadhu MJ, **Durvasula A**, Bloom JS, Kruglyak L. Ancient balancing selection maintains incompatible versions of the galactose pathway in yeast (2021). *Science*.
 - Highlighted in *Science*
9. PopSim Consortium. A community-maintained standard library of population genetic models (2020) *eLife*.
 - Highlighted in *Nature Methods*
8. **Durvasula A**, Sankararaman S. Recovering signals of ghost archaic admixture in the genomes of present-day Africans (2020). *Science Advances*.
 - F1000 recommended, highlighted in: New York Times, NPR, BBC, other major news outlets
7. **Durvasula A**, Sankararaman S. A statistical model for reference-free inference of local archaic ancestry (2019). *PLoS Genetics*.
6. Huber CD*, **Durvasula A***, Hancock AM, Lohmueller KE. Gene expression drives the evolution of dominance (2018). *Nature Communications*.

5. Schumer M, Xu C, Powell D, **Durvasula A**, Skov L, Holland C, Sankararaman S, Andolfatto P, Rosenthal G, Przeworski M. Natural selection interacts with the local recombination rate to shape the evolution of hybrid genomes (2018). *Science*.
► F1000 recommended
4. Schweizer RM, **Durvasula A**, Smith J, Vohr SH, Stahler DR, Galaverni M, Thalmann O, Smith D, Randi E, Green RE, Lohmueller KE, Novembre J, Wayne RK. The evolutionary history of a selectively swept coat color and immunity locus in North American wolves (2018). *Molecular Biology and Evolution*.
3. **Durvasula A***, Fulgione A*, Gutaker RM, Alacakaptan SI, Flood PJ, Neto C, Alonso-Blanco C, Burbano HA, Pico FX, Tsuchimatsu T, Hancock AM. African genomes illuminate the early history and transition to selfing in *Arabidopsis thaliana* (2017). *Proceedings of the National Academy of Sciences*.
► F1000 recommended, highlighted in: *Science, Nature Plants*
2. **Durvasula A***, Hoffman PJ*, Kent TV, Liu C, Kono TJY, Morrell PL, Ross-Ibarra J. angsd-wrapper: utilities for analysing next-generation sequencing data (2016) *Mol Ecol Resour*.
1. Beissinger T, Wang L, Crosby K, **Durvasula A**, Hufford M, Ross-Ibarra J. Recent demography drives changes in linked selection across the maize genome (2016). *Nature Plants*.

Grants and Awards

Poster Prize: EMBL Reconstructing the human past: using ancient and modern genomics	2022
ASHG Reviewer's Choice Abstract (top 10%)	2022
Dimitris N. Chorafas Foundation Award	2021
Graduate Research Fellowship Program, National Science Foundation	2018-2021
International Conference for Quantitative Genetics Early Career Researcher Scholarship	2020
Stephen D. Cederbaum Travel Award, UCLA Genetics and Genomics	2019
Stephen D. Cederbaum Travel Award, UCLA Genetics and Genomics	2018
Kenneth I. Shine Fellowship, UCLA	2018
Vienna Biocenter Summer Internship Scholarship	2015
Undergraduate Travel Award, UC Davis Plant Sciences	2015

Invited Seminars

Center for Computational Molecular Biology, Brown University, Providence, Rhode Island	2021
Analytical and Translational Genetics Unit, Massachusetts General Hospital	2021
Evolutionary Genetics Super-group Meeting, University of Chicago, Illinois	2021
Department of Ecology and Evolutionary Biology, University of Toronto, Canada	2021
Salk Institute, La Jolla, California	2021
Rasmus Nielsen Lab, UC Berkeley, California	2020
Human Genetics Seminar Series, UCLA, California	2018

Conference talks

Population, Evolutionary, and Quatitative Genetics. Virtual	2020
Human Evolution. Hinxton, United Kingdom	2019
Society for Molecular Biology and Evolution. Yokohama, Japan	2018
RECOMB-Genetics. Los Angeles, California	2017
Vienna Biocenter Summer Symposium. Vienna, Austria	2015

Teaching and Service

Guest lecture: The neutral theory of evolution (UCLA), Spring 2019

Teaching assistant: Population Genetics (Undergraduate level course; UCLA), Spring 2018, 2019

Teaching assistant: Ecological Genomics (Graduate level course; UC Davis), Winter 2015

Mentoring:

Roslyn Curry (post-bac, Harvard)

Amar Sarkar (graduate student, Harvard)

Scott Shi (undergraduate, UCLA)

Mario Paciuc (Bruins in Genomics, UCLA)

Elliot Kang (Bruins in Genomics, UCLA)

Peer review: Bioinformatics; G3: Genes, Genomes, Genetics; Genome Biology and Evolution; Human Genetics and Genomics Advances; International Society for Computational Biology; Molecular Ecology Resources; Nature Ecology and Evolution; PCI Evolutionary Biology, Philosophical Transactions of the Royal Society: B; PLoS Genetics

Software

ANGSD-wrapper: <https://github.com/ANGSD-wrapper/angsd-wrapper>

ArchIE: <https://github.com/sriramlab/ArchIE>

OrientAGraph: <https://github.com/ekmolloy/OrientAGraph>

stdpopsim: <https://github.com/popsim-consortium/stdpopsim>